Requirements Analysis Report

Project Title: Development of Integrated E- Library Management System

Client Organization: Ethiopian Geological Institute (EGI)

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1. Introduction

The Ethiopian Geological Institute (EGI) requires a scalable digital platform for managing geological reports, GIS datasets, laboratory records, research documents, and administrative information in a centralized environment. The proposed system aims to improve data accessibility, security, information sharing, analytics, and institutional decision-making.

This Requirements Analysis Report defines the functional and non-functional requirements of the proposed Geological Information Management System, including technological requirements, database specifications, user management, security, and additional digital services identified during the inception phase.

2. Project Objectives

The proposed system is intended to achieve the following objectives:

- Develop a centralized digital repository for geological information.
- Improve institutional data accessibility and sharing.
- Support secure storage and preservation of digital resources.
- Provide advanced search and GIS-based retrieval mechanisms.
- Enable online access and digital service delivery.
- Support commercial distribution of premium geological reports.
- Improve operational efficiency and institutional collaboration.
- Support analytics and decision-making processes.

3. Functional Requirements

Functional requirements describe the core services and operations the system must provide.

3.1 User Authentication and Authorization

The system should provide secure authentication and authorization services.

Required Features

- User registration and login
- Password encryption and recovery
- Role-based access control (RBAC)
- Multi-level user permissions
- Session management
- Activity logging

The authentication mechanism should ensure secure access to institutional and public resources.

3.2 Document and Data Management

The platform should support efficient management of geological reports and institutional datasets.

Required Features

- Document upload and download
- Bulk file upload
- Metadata management
- File categorization and classification
- Version control
- Document approval workflow
- File preview capability
- Multimedia management

The system should support multiple file formats including PDF, GIS layers, Excel files, images, and multimedia resources.

3.3 Geological Metadata Management

The institution requires a flexible metadata structure capable of supporting geological datasets.

Required Features

- Custom geological metadata schema
- Spatial metadata support
- Metadata validation
- Metadata editing and updating
- Advanced metadata filtering
- Metadata indexing for search optimization

The metadata model should support geological reports, borehole logs, GIS datasets, mineral exploration records, and laboratory information.

3.4 Search and Retrieval Services

The system should provide advanced and high-performance search functionality.

Required Features

- Full-text search
- Metadata-based search
- GIS/spatial search
- Advanced filtering options
- Elasticsearch integration
- Search suggestions and indexing
- Search result categorization

The search engine should support fast retrieval of both structured and unstructured datasets.

3.5 GIS and Spatial Services

GIS integration is a core institutional requirement.

Required Features

- Spatial data upload
- GIS layer management
- Interactive map visualization
- Coordinate-based search
- GeoServer integration
- Spatial metadata support
- GIS analytics and overlays

The platform should support geological maps and spatial analysis services.

3.6 Payment and Commercial Services

The institution requires mechanisms for monetizing premium geological resources.

Required Features

- Online report purchasing
- Secure digital downloads
- Telebirr integration
- Chapa payment integration
- Transaction tracking
- Invoice generation
- Subscription and pricing management

The payment system should provide secure and transparent commercial services.

3.7 Reporting and Analytics

The system should support analytics and decision-making services.

Required Features

- Dashboard visualization
- Usage statistics
- Download tracking
- User activity reports
- Metadata analytics
- GIS-based analytics
- Exportable reports

Management dashboards should support operational monitoring and institutional planning.

4. Non-Functional Requirements

Non-functional requirements define system quality, performance, and operational standards.

4.1 Performance Requirements

The system should:

- Support fast response times
- Handle large datasets efficiently
- Support concurrent users
- Provide scalable architecture
- Support optimized indexing and retrieval

Elasticsearch integration will improve search performance for large geological repositories.

4.2 Security Requirements

Geological information requires strong security controls.

Security Features

- JWT-based authentication
- Role-based access control
- HTTPS encrypted communication
- Password hashing and encryption

- Audit logging
- Backup and disaster recovery
- Secure API communication
- File access restrictions

The platform should protect sensitive institutional and commercial information.

4.3 Usability Requirements

The system should provide a modern and user-friendly environment.

Usability Features

- Responsive web interface
- Mobile accessibility
- Simple navigation structure
- Interactive dashboards
- User-friendly search interface
- Accessibility support

A modern React-based frontend will improve user experience and staff productivity.

4.4 Reliability and Availability

The platform should ensure continuous and stable service delivery.

Reliability Features

- Automated backup systems
- Error recovery mechanisms
- Fault-tolerant architecture
- Cloud-ready deployment
- Monitoring and logging services

The system should minimize downtime and support long-term operational continuity.

5. Technological Requirements

The proposed system requires modern and scalable technologies.

Component	Technology
Frontend	React.js

Backend	Node.js
database	PostgreSQL
Search Engine	Search Engine
Authentication	JWT / OAuth
Payment Integration	Tellbirr/ CBE
Hosting Environment	Linux Server / Cloud
Version Control	Git/GitHub

6. Database Specifications

The proposed platform requires a centralized relational database architecture capable of supporting structured and spatial datasets.

Database Requirements

- PostgreSQL relational database
- GIS spatial database support
- Metadata indexing
- Relational data modeling
- Scalable architecture
- Backup and recovery mechanisms
- Data integrity constraints

Core Database Tables

The database should support future expansion and integration with external systems.

7. User Management Structure

The system should support multiple user roles with different permissions.

Table	Purpose
users	User accounts
Roles	Permissions
Documents	Report metadata
Categories	Resource classification
gis_data	GIS datasets
uploads	Upload tracking
payments	Payment records

downloads	Download history
audit_logs	Activity logs
notifications	System alerts

Public User

- Browse metadata
- Search public reports
- Purchase premium resources
- Download approved public documents

Registered Researcher

- Access assigned datasets
- Download purchased reports
- Access research resources
- Manage personal activity history

Data Validator (Staff)

- Review uploaded records
- Validate metadata quality
- Approve or reject submissions
- Maintain institutional data standards

System Administrator

- Manage users and permissions
- Configure payment systems
- Monitor system performance
- Manage security and backups

8. Proposed Additional Services

To improve institutional efficiency and future scalability, the following additional services are recommended.

API Integration Services

Support interoperability with external government and research systems.

AI-Assisted Search

Future integration of intelligent search and metadata recommendation services.

Mobile Accessibility

Responsive access through smartphones and tablets.

Notification Services

Email and system notifications for uploads, approvals, and transactions.

Digital Preservation Services

Long-term archival support for institutional geological records.

9. Implementation Approach

The project will follow Agile Scrum methodology to support iterative development and continuous stakeholder feedback.

Advantages of Agile Approach

- Early MVP delivery
- Continuous stakeholder participation
- Faster issue identification
- Incremental feature improvements
- Reduced development risks

The methodology will ensure flexibility throughout implementation.

10. Conclusion

The Requirements Analysis confirms the need for a centralized, secure, scalable, and modern Geological Information Management System capable of supporting data ingestion, storage, management, GIS integration, analytics, and commercial digital services.

The proposed architecture and functional requirements provide a strong technical foundation for developing a future-ready institutional platform that improves operational efficiency, collaboration, information accessibility, and evidence-based decision-making within the Ethiopian Geological Institute.